Homework 7

STAT 300 (Fall 2024)

Clearly label your answers to each question and each sub-question. Your answers MUST be uploaded to Canvas as a <HWx_Yourfirtname.nb.html> file by the deadline.

(1) Suppose we have data that includes information on movies that came out of Hollywood in 2011. We want to build a model to predict *profitability*, which is the percent of the budget recovered in profits. The dataset contains five **explanatory** variables and the details are as follows.

Variable Name	Description
RottenTomatoes	Meata rating of critical reviews, from the Rotten Tomatoes website
AudienceScore	Average audience score, from the Rotten Tomatoes website
TheatersOpenWeek	Number of theaters showing the moview on opening weekend
BOAverageOpenWeek	Average box office revenue per heater opening week-end, in dollars
DomesticGross	Gross revenus in the US by the end of 2011, in millions of dollars

Use the $Adj - R^2$ value in forward model selection approach to fit the **best** first order model for this data. Clearly state an expression of the model at each step until the variable selection stops. When the variable selection is stopped, give an *expression* of variables in the model.

Variable(s)	R^2	Adj. R^2
RottenTomatoes	0.024	0.015
AudienceScore	0.010	0.001
TheatersOpenWeek	0.006	- 0.002
BOAverageOpenWeek	0.008	-0.001
DomesticGross	0.036	0.028
DomesticGross, RottenTomatoes	0.049	0.033
DomesticGross, AudienceScore	0.037	0.021
DomesticGross, TheatersOpenWeek	0.038	0.021
DomesticGross, BOAverageOpenWeek	0.036	0.021
DomesticGross, RottenTomatoes, AudienceScore	0.061	0.037
DomesticGross, RottenTomatoes, TheatersOpenWeek	0.050	0.025
DomesticGross, RottenTomatoes, BOAverageOpenWeek	0.050	0.025
DomesticGross, RottenTomatoes, AudienceScore, TheatersOpenWeek	0.063	0.029
DomesticGross, RottenTomatoes, AudienceScore, BOAverageOpenWeek	0.062	0.028

(2) A research group collected data from a random sample of women who were having trouble getting pregnant. Data is in *Fertility.csv* file. Variable description is as follows:

Age - Age in years MeanAFC - Average antral follicle count FSH - Maximum follicle stimulating hormone level E2 - Fertility level MaxE2 - Maximum fertility level MaxDailyGn - Maximum daily gonadotropin level TotalGn - Total gonadotropin level Oocytes - Number of egg cells Embryos - Number of embryos

The research group is interested in developing a model to predict the MeanAFC.

(a) If you were to use the best subset approach to select a model, how many potential models should you check in the process?

(b) Use the AIC in bi-directional selection to obtain a model.

(c) Fit the model with the predictors you selected in part b. Write down the equation of the fitted regression model.

(d) Check the validity of the assumptions for the model you pick in part b.

(e) What is the coefficient of determination for the model chosen in part b?

(f) Now use BIC in bi-directional selection to obtain a model.

(g) Fit the model with the predictors you selected in part f. Write down the equation of the fitted regression model.

(h) Check the validity of the assumptions for the model you pick in part f.

(i) What is the coefficient of determination for the model chosen in part f?

(j) To see if the model chosen is the same with step wise selection with BIC, use best subset approach with BIC. Do you still get the same model?

(k) Check the possible multicollinearity of the above chosen models. Comment on them.

(l) Using the models you get from the above different approaches recommend a final model. When you provide recommendations consider, R^2 values, validity of the assumptions, prediction accuracy, multicollinearity.